
p -ADIC HAHN SERIES WITH SPARSE SUPPORT

by

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1. Introduction

Let p be a prime number. The p -adic Hahn series are generalized formal power series of the form $f = \sum_{q \in \mathbf{Q}} [f(q)]p^q$, where $f(q) \in \overline{\mathbf{F}}_p$ and $[\cdot]$ is the Teichmüller lift, whose support $\text{Supp}(f) = \{q \in \mathbf{Q} \mid f(q) \neq 0\}$ is a well-ordered subset of $\mathbf{Q}$. Krull, Lampert, and Poonen showed that the set of p -adic Hahn series, which we denote by $\mathbf{L}_p$, forms the spherical completion of $\overline{\mathbf{Q}}_p$, the field of p -adic algebraic numbers. In particular, $\mathbf{L}_p$ is algebraically closed and complete with respect to the p -adic valuation given by $f \mapsto \min \text{Supp}(f)$.

The field $\mathbf{L}_p$ provides a natural setting to study transcendental number theory over p -adic fields. More precisely, the following question is of fundamental importance:

Question 1.1. — *Given a p -adic Hahn series $f \in \mathbf{L}_p$, how can one determine whether f is algebraic over $\overline{\mathbf{Q}}_p$?*

Although this question remains open, several necessary conditions for a p -adic Hahn series to be algebraic over $\overline{\mathbf{Q}}_p$ are known:

1. In [Lam86] and [Poo93], Lampert and Poonen proved that if $f \in \mathbf{L}_p$ is algebraic over $\overline{\mathbf{Q}}_p$, then
 - (a) there exists an integer T such that $\text{Supp}(f) \subset \frac{1}{T}\mathbf{Z}[1/p]$;
 - (b) there exists a finite extension $\mathbf{F}_q$ of $\mathbf{F}_p$ such that $\{f(q)\}_{q \in \mathbf{Q}} \subset \mathbf{F}_q$.

These conditions are also studied quantitatively in [WY24].

2. In [Lam86], Lampert also proved that if $f \in \mathbf{L}_p$ is algebraic over $\overline{\mathbf{Q}}_p$, then the accumulation points of $\text{Supp}(f)$ are rational numbers.

3. In [Ked01] and [Ked17], Kedlaya gives a necessary and sufficient condition for an equal-characteristic Hahn series in $\mathbf{L}_p^b := \overline{\mathbf{F}}_p((t^{\mathbf{Q}}))$ to lie in the algebraic closure $\overline{\mathbf{F}}_p((t))^{\text{alg}}$ of $\overline{\mathbf{F}}_p((t))$, phrased in the language of automata theory. As an application, Kedlaya uses Witt vectors to lift this result to the p -adic case: he shows that the field $\mathbf{C}_p := \widehat{\mathbf{Q}}_p$ of p -adic complex numbers, when viewed as a subfield of $\mathbf{L}_p$, coincides with the completion of the set $\Theta(\overline{\mathbf{F}}_p((t))^{\text{alg}, \wedge})$, where $\overline{\mathbf{F}}_p((t))^{\text{alg}, \wedge}$ is the t -adic completion of $\overline{\mathbf{F}}_p((t))^{\text{alg}}$ and $\Theta: \overline{\mathbf{F}}_p((t^{\mathbf{Q}})) \rightarrow \mathbf{L}_p$ is the map $\sum_{q \in \mathbf{Q}} f(q)t^q \mapsto \sum_{q \in \mathbf{Q}} [f(q)]p^q$.

Intuitively, Kedlaya's result indicates that the algebraicity of equal-characteristic Hahn series in $\overline{\mathbf{F}}_p((t^{\mathbf{Q}}))$ and the algebraicity of p -adic Hahn series in $\mathbf{L}_p$ are related to some extent via the map Θ , which leads to the following questions:

Question 1.2. —

1. Suppose that $f \in \mathbf{L}_p^b$ is algebraic over $\mathbf{F}_p((t))$. Is $\Theta(f)$ algebraic over $\mathbf{Q}_p$?
2. Suppose that $f \in \mathbf{L}_p$ is algebraic over $\mathbf{Q}_p$. Is $\Theta^{-1}(f)$ algebraic over $\mathbf{F}_p((t))$?

In [WY25], we gave a negative answer to the first question by showing that the p -adic Hahn series $\mathfrak{A} := \sum_{k=1}^{\infty} p^{-1/p^k}$ is transcendental over $\mathbf{Q}_p$, whereas its preimage under Θ is a root of the polynomial $X^p - X - t^{-1}$ over $\mathbf{F}_p((t))$. The key observation is that the support of $\mathfrak{A}$ is “sparse”, in the sense that for any non-zero polynomial $P(X) \in \mathbf{Q}_p[X]$, the multinomial expansion of $P(\mathfrak{A})$ contains orphaned terms that cannot be cancelled by other terms, which forces $P(\mathfrak{A})$ to be nonzero. In this article, we generalize this idea to the following combinatorial definition:

Definition 1.3. — Let $S \subsetneq [0, 1)$ be a set consisting of finite decimals in base p with integral part zero⁽¹⁾. We say that S is *sparse* if:

1. there exists an integer $c \geq 1$ such that the digit sum of every element of S is at most c ;
2. for infinitely many integers $n \geq 1$, there exist n elements $d_1, \dots, d_n \in S$ such that
 - (a) the digit sum of each d_i is exactly c ;
 - (b) there is no carry when adding $d_1, \dots, d_n$ together;
 - (c) if $e_1, \dots, e_n \in S$ satisfy that $d_1 + \dots + d_n$ and $e_1 + \dots + e_n$ differ by an integer, then up to a permutation of $\{1, 2, \dots, n\}$, $d_i = e_i$ for every $i = 1, 2, \dots, n$.

Remark 1.4. — The first condition in Definition 1.3 is closely related to Kedlaya's criterion for the algebraicity of equal-characteristic Hahn series: he shows that if $f \in \mathbf{L}_p^b$ is algebraic over $\overline{\mathbf{F}}_p((t))$, then there exist integers $c, T \geq 1$ such that the digit sums of the decimal part of elements of $-T \cdot \text{Supp}(f)$ are uniformly bounded by c .

Theorem 1.5 (cf. Theorem 5.1). — Let $f \in \mathbf{L}_p$ be a p -adic Hahn series such that there exists an integer T for which $-T \cdot \text{Supp}(f)$ admits a sparse set of representatives modulo $\mathbf{Z}$. Then f is transcendental over $\widehat{\mathbf{Q}}_p$, and hence over $\mathbf{Q}_p$.

Remark 1.6. — The sparseness condition says nothing about the coefficients of f , so it is reasonable that it does not distinguish algebraicity over $\mathbf{Q}_p$ from algebraicity over $\widehat{\mathbf{Q}}_p$.

2. Preliminaries on Hahn series

To keep this article self-contained, we briefly recall some basic facts about Hahn series.

Definition 2.1 ([Poo93, Section 3]). — Let R be a commutative ring and G be an ordered group.

⁽¹⁾These elements have the form $\sum_{i=1}^{\infty} a_i p^i$ with $a_i \in \{0, 1, \dots, p-1\}$ and $a_i = 0$ for i large enough.

1. For any $f \in \text{Hom}_{\text{Set}}(G, R)$, we define the **support** of f to be

$$\text{Supp}(f) = \{g \in G : f(g) \neq 0\}.$$

2. Define the set of **Hahn series** over R with value group G to be

$$R((G)) := \{f \in \text{Hom}_{\text{Set}}(G, R) : \text{Supp}(f) \text{ is well-ordered}\}.$$

By introducing a formal variable t , elements in $R((G))$ will also be written as $\sum_{g \in G} r_g t^g$, where $r_g \in R$ for all $g \in G$.

Proposition 2.2 ([Poo93, Lemma 1, Corollary 2]). — Let R be a commutative ring and G be an ordered group.

1. With identity $1 \cdot t^0$ and addition as well as multiplication given by

$$\sum_{g \in G} a_g t^g + \sum_{g \in G} b_g t^g := \sum_{g \in G} (a_g + b_g) t^g, \quad \sum_{g \in G} a_g t^g \cdot \sum_{g \in G} b_g t^g := \sum_{g \in G} \left(\sum_{h \in G} a_h b_{g-h} \right) t^g$$

$R((G))$ forms a commutative ring.

2. If R is a field, then so is $R((G))$. Moreover, with the map

$$v : R((G)) \longrightarrow G \cup \{\infty\}, \quad f \longmapsto \begin{cases} \min \text{Supp}(f), & \text{if } f \neq 0 \\ \infty, & \text{if } f = 0 \end{cases}$$

$R((G))$ becomes a valued field with value group G and residue field R .

Note that $\text{char } R((G)) = \text{char } R$, we call $R((G))$ the **equal-characteristic field of Hahn series** over R with value group G , also denoted by $R((t^G))$ with respect to the formal variable t .

Proposition 2.3 ([Poo93, Proposition 3, Corollary 3, Proposition 5])

Let k be a perfect field of characteristic p and G be an ordered group containing $\mathbb{Z}$ as a subgroup. Besides that, let

$$\mathcal{N} := \left\{ \sum_{g \in G} r_g t^g \in W(k)((t^G)) : \text{for every } g \in G, \sum_{n \in \mathbb{Z}} r_{g+n} p^n = 0 \right\},$$

where $W(k)$ is the ring of Witt vectors of k . We call elements in $\mathcal{N}$ the **null series** of $W(k)((t^G))$. Then

1. $\mathcal{N}$ is a maximal ideal of $W(k)((t^G))$, which makes $W(k)((p^G)) := W(k)((t^G))/\mathcal{N}$ a field⁽²⁾, called the **p -adic field of Hahn series**.
2. Every element in $W(k)((p^G))$ can be uniquely written as

$$\sum_{g \in G} [r_g] p^g,$$

where $r_g \in k$ for all $g \in G$ and $[\cdot] : k \longrightarrow W(k)$ is the Teichmüller lift. We call this the **standard expansion** of the element.

3. For $f = \sum_{g \in G} [r_g] p^g$, define the **support** of f to be

$$\text{Supp}(f) = \{g \in G : r_g \neq 0\}.$$

Then the map

$$v : W(k)((G))/\mathcal{N} \longrightarrow G \cup \{\infty\}, \quad f \mapsto \begin{cases} \min \text{Supp}(f), & \text{if } f \neq 0 \\ \infty, & \text{if } f = 0 \end{cases}$$

⁽²⁾Intuitively speaking, $W(k)((p^G))$ is obtained by replacing the formal variable t of elements in $W(k)((t^G))$ by the prime p .

makes $W(k)((G))/\mathcal{N}$ a mixed-characteristic valued field with value group G and residue field k .

The most important property of the field of Hahn series is the following:

Theorem 2.4 (cf. [Poo93, Theorem 1, Corollary 4, Corollary 6])

Let F be an equal-characteristic (resp. mixed-characteristic) valued field with divisible value group G and algebraically closed residue field k . Then the equal-characteristic (resp. p -adic) field of Hahn series $k((t^G))$ (resp. $W(k)((p^G))$) is the unique (up to isomorphisms of valued field) minimal spherically complete extension of F . Moreover, it is algebraically closed and complete.

The following examples are the fields of Hahn series used in this article:

Example 2.5. — Let $F = \bar{\mathbf{F}}_p((t))$ (resp. $\check{\mathbf{Q}}_p = W(\bar{\mathbf{F}}_p)[p^{-1}]$), which has value group $\mathbf{Q}$ and residue field $\bar{\mathbf{F}}_p$. Then the field of equal-characteristic (resp. p -adic) Hahn series $\mathbf{L}_p^b := \bar{\mathbf{F}}_p((t^{\mathbf{Q}}))$ (resp. $\mathbf{L}_p := \mathcal{O}_{\check{\mathbf{Q}}_p}((p^{\mathbf{Q}})) = W(\bar{\mathbf{F}}_p)((p^{\mathbf{Q}}))$) is the spherical completion of F with the same residue field and value group, and is algebraically closed and complete.

Remark 2.6. — To simplify the statement, we will simply call $\mathcal{O}_{\check{\mathbf{Q}}_p}((p^{\mathbf{Q}}))$ the p -adic Hahn series without specifying the residue field and value group.

3. Combinatorial conditions

In this section, we give a rigorous formulation of the sparseness condition (cf. Definition 1.3) and establish some of its properties. These properties will be used in the proof of Theorem 1.5.

Definition 3.1. —

1. For $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, let $\Sigma(\underline{d}) := \sum_{i=1}^{\infty} d_i$ and $\|\underline{d}\| := \sum_{i=1}^{\infty} d_i p^{-i}$.
2. Let $\mathbb{P} := \bigoplus_{\mathbf{Z}_{\geq 1}} \{0, 1, \dots, p-1\} \subsetneq \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$.

Lemma 3.2. — For any $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, there exists a unique element $\tau(\underline{d}) \in \mathbb{P}$ such that $\|\underline{d}\| - \|\tau(\underline{d})\| \in \mathbf{Z}$.

Proof. — If one writes $\|\underline{d}\|$ as a decimal expansion⁽³⁾ in base p :

$$\|\underline{d}\| = w.d_1 \cdots d_n \cdots,$$

where $w \in \mathbf{Z}$ and $d_i \in \{0, 1, \dots, p-1\}$ for any i , then $\tau(\underline{d}) = 0.d_1 \cdots d_n \cdots$. The uniqueness is trivial. $\square$

We collect several properties of these concepts in the following lemma:

Lemma 3.3. — Let $\underline{d}, \underline{e} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$.

1. The maps $\Sigma(\cdot)$ and $\|\cdot\|$ are linear. Moreover, $\|\cdot\|$ is injective when restricted to $\mathbb{P}$.
2. One has $\tau(\underline{d}) = \tau(\underline{e})$ if and only if $\|\underline{d}\| - \|\underline{e}\| \in \mathbf{Z}$.
3. $\underline{d} \in \mathbb{P}$ if and only if $\tau(\underline{d}) = \underline{d}$.
4. One has $\Sigma(\tau(\underline{d})) \leq \Sigma(\underline{d})$, with equality if and only if $\underline{d} \in \mathbb{P}$.

Proof. — The first three statements are straightforward. We only prove the last statement.

For any $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, let

$$N(\underline{d}) := \begin{cases} 0, & \text{if } \underline{d} \in \mathbb{P}; \\ \max\{i \mid d_i \geq p\}, & \text{otherwise.} \end{cases}$$

By a descent argument on $N(\underline{d})$, it suffices to show that if $N(\underline{d}) \geq 1$, then there exists $\underline{e} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$ such that $\|\underline{d}\| - \|\underline{e}\| \in \mathbf{Z}$, $\Sigma(\underline{e}) < \Sigma(\underline{d})$, and $N(\underline{e}) < N(\underline{d})$.

⁽³⁾We do not allow infinite strings of $(p-1)$ in the decimal expansion of $\|\underline{d}\|$ to ensure the uniqueness of $\tau(\underline{d})$.

Writing $d_{N(\underline{d})} = p \cdot r + s$ with $r \in \mathbf{N}$ and $s \in \{0, 1, \dots, p-1\}$, we take $\underline{e} := (d_0, \dots, d_{N(\underline{d})-2}, d_{N(\underline{d})-1} + r, s, 0, \dots)$. Then

$$\Sigma(\underline{d}) - \Sigma(\underline{e}) = d_{N(\underline{d})} - r - s = (p-1) \cdot r > 0$$

and $\|\underline{d}\| - \|\underline{e}\| = 0$. The result follows. $\square$

Definition 3.4. — Let p be a prime number.

1. Let $c, n \geq 1$ be integers. A subset $S \subset \mathbb{P}$ is (c, n) -**sparse** if
 - (a) there exists $c \geq 1$ such that $\Sigma(\underline{d}) \leq c$ for every $\underline{d} \in S$;
 - (b) there exist n (not necessarily distinct) elements $\underline{d}^{(1)}, \underline{d}^{(2)}, \dots, \underline{d}^{(n)} \in S$ such that
 - (i) $\Sigma(\underline{d}^{(i)}) = c$ for $i = 1, 2, \dots, n$ and $\sum_{i=1}^n d_j^{(i)} < p$ for $j \in \mathbf{N}$ (that is, $\sum_{i=1}^n \underline{d}^{(i)} \in \mathbb{P}$);
 - (ii) if $\underline{e}^{(1)}, \underline{e}^{(2)}, \dots, \underline{e}^{(n)} \in S$ satisfy $\left\| \sum_{i=1}^n \underline{d}^{(i)} \right\| - \left\| \sum_{i=1}^n \underline{e}^{(i)} \right\| \in \mathbf{Z}$, then up to a permutation of $\{1, 2, \dots, n\}$, $\underline{d}^{(i)} = \underline{e}^{(i)}$ for every $i = 1, 2, \dots, n$.
2. A subset $S \subset \mathbb{P}$ is **sparse** if there exists an integer $c \geq 1$ such that S is (c, n) -sparse for infinitely many integers $n \geq 1$.

Lemma 3.5. — Let $S \subset \mathbb{P}$ be a (c, n) -sparse subset for some integers $c, n \geq 1$. Let

$$\phi_0: S \longrightarrow \mathbf{N}, \underline{d} \longmapsto |\{i | \underline{d} = \underline{d}^{(i)}\}|.$$

Then ϕ_0 is the unique function $\phi: S \rightarrow \mathbf{N}$ such that $\sum_{\underline{d} \in S} \phi(\underline{d}) \leq n$ and

$$\sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi(\underline{d}) \equiv \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}) \pmod{\mathbf{Z}}.$$

Proof. — Suppose that $\phi_1: S \rightarrow \mathbf{N}$ is another function such that $\sum_{\underline{d} \in S} \phi_1(\underline{d}) \leq n$ and

$$\sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_1(\underline{d}) \equiv \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}) \pmod{\mathbf{Z}}.$$

1. If $\sum_{\underline{d} \in S} \phi_1(\underline{d}) < n$, then by Lemma 3.3 (4) we have

$$\Sigma \left(\tau \left(\sum_{\underline{d} \in S} \underline{d} \cdot \phi_1(\underline{d}) \right) \right) \leq \Sigma \left(\sum_{\underline{d} \in S} \phi_1(\underline{d}) \cdot \underline{d} \right) = \sum_{\underline{d} \in S} \phi_1(\underline{d}) \sum_{i=1}^{\infty} \underline{d}_i < n \cdot c.$$

On the other hand, the sparse condition (2a) of Definition 3.4 implies that $\sum_{j=1}^n d_i^{(j)} < p$ for any i , hence

$$\sum_{\underline{d} \in S} \underline{d} \cdot \phi_0(\underline{d}) = \left(\sum_{j=1}^n d_i^{(j)} \right)_{i \in \mathbf{Z}_{\geq 0}} = \tau \left(\sum_{\underline{d} \in S} \underline{d} \cdot \phi_0(\underline{d}) \right)$$

and consequently

$$\Sigma \left(\tau \left(\sum_{\underline{d} \in S} \underline{d} \cdot \phi_0(\underline{d}) \right) \right) = \sum_{i=0}^{\infty} \left(\sum_{j=1}^n d_i^{(j)} \right) = \sum_{j=1}^n \Sigma(d^{(j)}) = n \cdot c.$$

This leads to a contradiction by Lemma 3.3 (2).

2. If $\sum_{\underline{d} \in S} \phi_1(\underline{d}) = n$, then we take $\underline{e}^{(1)}, \underline{e}^{(2)}, \dots, \underline{e}^{(n)} \in S$ such that $\phi_1(\underline{d}) = |\{i | \underline{d} = \underline{e}^{(i)}\}|$ for any $\underline{d} \in S$. Then

$$\sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_1(\underline{d}) - \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}) = \left\| \sum_{\underline{d} \in S} \underline{d} \cdot \phi_1(\underline{d}) \right\| - \left\| \sum_{\underline{d} \in S} \underline{d} \cdot \phi_0(\underline{d}) \right\| = \left\| \sum_{i=1}^n \underline{e}^{(i)} \right\| - \left\| \sum_{i=1}^n \underline{d}^{(i)} \right\| \in \mathbf{Z}.$$

By the sparse condition (2b) of Definition 3.4, up to a permutation of $\{1, 2, \dots, n\}$, $\underline{d}^{(i)} = \underline{e}^{(i)}$ for any $i = 1, 2, \dots, n$. This implies that $\phi_1 = \phi_0$. $\square$

4. T -scaled realization of $\mathbf{L}_p$

To prove Theorem 1.5, one needs to group the terms of a p -adic Hahn series $f = \sum_{q \in \mathbf{Q}} [f(q)]p^q$ in $\mathbf{L}_p$ with $\text{Supp}(f) \subset \frac{1}{T}\mathbf{Z}[1/p]$ by the residue of the exponent modulo $\frac{1}{T}\mathbf{Z}$ for some integer $T \geq 1$. This works when $T = 1$, since a direct computation shows that the element

$$\sum_{q \in \text{Supp}(f)/\mathbf{Z}} \left(\sum_{w \in \mathbf{Z}} f(q+w)p^w \right) t^q \in W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}}))$$

is a preimage of f under the natural projection $W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}})) \rightarrow W(\overline{\mathbf{F}}_p)((p^{\mathbf{Q}}))$, where

$$\text{Supp}(f)/\mathbf{Z} := \{\inf(\text{Supp}(f) \cap (q + \mathbf{Z})) | q \in \text{Supp}(f)\} \subset \mathbf{Q}$$

is a well-ordered set of representatives of $\text{Supp}(f)$ modulo $\mathbf{Z}$. However, when $T > 1$, the same construction fails, because the element $\sum_{w \in \frac{1}{T}\mathbf{Z}} f(q+w)p^w$ does not necessarily lie in $W(\overline{\mathbf{F}}_p)$, so a direct analogue of the above construction is not well-defined⁽⁴⁾. To resolve this, we enlarge the ring of Witt vectors $W(\overline{\mathbf{F}}_p)$ to include $p^{1/T}$, and realize $\mathbf{L}_p$ as a quotient of $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$ by a suitable ideal. This is the main content of this section.

Throughout the rest of this section, T denotes a positive integer.

Lemma 4.1. — *One has $[\check{\mathbf{Q}}_p(p^{1/T}) : \check{\mathbf{Q}}_p] = T$. In particular, $1, p^{1/T}, \dots, p^{(T-1)/T}$ form a basis of $\check{\mathbf{Q}}_p(p^{1/T})$ over $\check{\mathbf{Q}}_p$.*

Proof. — Since $\check{\mathbf{Q}}_p$ is a discrete valuation field and $p^{1/T}$ is a root of the Eisenstein polynomial $X^T - p$, the result follows from the Eisenstein criterion. $\square$

Remark 4.2. — $W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}}))$ is a subfield of $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$ via the natural inclusion $W(\overline{\mathbf{F}}_p) \subset W(\overline{\mathbf{F}}_p)[p^{1/T}]$.

Definition 4.3. — *Let $\mathcal{N}_T$ be the set of elements $\sum_{q \in \mathbf{Q}} c_q t^q \in W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$ that for any $q \in \mathbf{Q}$,*

$$\sum_{n \in \mathbf{Z}} c_{q+\frac{n}{T}} p^{\frac{n}{T}} = 0.$$

We call these elements the T -null-series in $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$.

Remark 4.4. — *Obviously, when $T = 1$, $\mathcal{N}_T$ coincides with the null-series in $W(\overline{\mathbf{F}}_p)((p^{\mathbf{Q}}))$ defined in Proposition 2.3.*

Lemma 4.5. —

1. $\mathcal{N}_T$ is an ideal of $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$.
2. For every element f of $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$, there exists a unique element $g = \sum_{q \in \mathbf{Q}} [g(q)]t^q$ in $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$ such that $f - g \in \mathcal{N}_T$.
3. $\mathcal{N}_T$ is a maximal ideal of $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$, so that $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T$ is a field.

Proof. — These three statements are generalizations of [Poo93, Proposition 3], [Poo93, Proposition 4] and [Poo93, Corollary 3] respectively, and the proofs are essentially the same. $\square$

⁽⁴⁾This was not noticed until we tried to formalize the proof of the main theorem in Lean 4.

Remark 4.6. — By (2) of this lemma, we will formally write elements of $W(\overline{\mathbf{F}}_p)[p^{1/T}](t^{\mathbf{Q}})/\mathcal{N}_T$ as $\sum_{q \in \mathbf{Q}} [g(q)]p^q$.

Lemma 4.7. — One has $\mathcal{N}_T \cap W(\overline{\mathbf{F}}_p)(t^{\mathbf{Q}}) = \mathcal{N}$.

Proof. — Let $f = \sum_{q \in \mathbf{Q}} [f(q)]t^q \in \mathcal{N}$, then for any $q \in \mathbf{Q}$,

$$(a) \quad \sum_{n \in \mathbf{Z}} [f(q+n)]p^n = 0.$$

Notice that for any $q \in \mathbf{Q}$, one has

$$\begin{aligned} \sum_{n \in \mathbf{Z}} \left[f\left(q + \frac{n}{T}\right) \right] p^{\frac{n}{T}} &= \sum_{u=0}^{T-1} \sum_{\substack{n \in \mathbf{Z} \\ n \equiv u \pmod{T}}} \left[f\left(q + \frac{n}{T}\right) \right] p^{\frac{n}{T}} \\ &= \sum_{u=0}^{T-1} p^{\frac{u}{T}} \sum_{n \in \mathbf{Z}} \left[f\left(q + \frac{u}{T} + n\right) \right] p^n. \end{aligned}$$

By applying (a) to $q + \frac{u}{T}$ for $u = 0, 1, \dots, T-1$, one has $\sum_{n \in \mathbf{Z}} [f(q + \frac{n}{T})]p^{\frac{n}{T}} = 0$. As a result, $f \in \mathcal{N}_T$.

Conversely, let $g = \sum_{q \in \mathbf{Q}} [g(q)]t^q \in \mathcal{N}_T$. Then for any $q \in \mathbf{Q}$, one has

$$\sum_{n \in \mathbf{Z}} \left[g\left(q + \frac{n}{T}\right) \right] p^{\frac{n}{T}} = \sum_{u=0}^{T-1} p^{\frac{u}{T}} \sum_{n \in \mathbf{Z}} \left[g\left(q + \frac{u}{T} + n\right) \right] p^n = 0.$$

By Lemma 4.1, for any $u = 0, 1, \dots, T-1$, one has $\sum_{n \in \mathbf{Z}} [g(q + \frac{u}{T} + n)]p^n = 0$. The result follows from taking $u = 0$. $\square$

Lemma 4.8. — One has $W(\overline{\mathbf{F}}_p)(t^{\mathbf{Q}}) + \mathcal{N}_T = W(\overline{\mathbf{F}}_p)[p^{1/T}](t^{\mathbf{Q}})$.

Proof. — Take $f = \sum_{q \in \mathbf{Q}} c_q t^q \in W(\overline{\mathbf{F}}_p)[p^{1/T}](t^{\mathbf{Q}})$, where

$$c_q = \sum_{n=0}^{\infty} [c_{q,n}]p^{\frac{n}{T}} \in W(\overline{\mathbf{F}}_p)[p^{1/T}].$$

Then one has $f = f_0 + p^{\frac{1}{T}} f_1 + \dots + p^{\frac{T-1}{T}} f_{T-1}$, where

$$f_u = \sum_{t \in \mathbf{Q}} \left(\sum_{\substack{n \in \mathbf{N} \\ n \equiv u \pmod{T}}} [c_{q,n}]p^{\frac{n-u}{T}} \right) t^q \in W(\overline{\mathbf{F}}_p)(t^{\mathbf{Q}}), \quad u = 0, 1, \dots, T-1.$$

It suffices to show that $p^{\frac{u}{T}} f_u$ belongs to $W(\overline{\mathbf{F}}_p)(t^{\mathbf{Q}}) + \mathcal{N}_T$ for $u = 0, 1, \dots, T-1$. This is clear if we write f_u as $f_u = (p^{\frac{u}{T}} \cdot t^0 - t^{\frac{u}{T}}) f_u + t^{\frac{u}{T}} f_u$, where $t^{\frac{u}{T}} f_u \in W(\overline{\mathbf{F}}_p)(t^{\mathbf{Q}})$ and $(p^{\frac{u}{T}} \cdot t^0 - t^{\frac{u}{T}}) f_u$ is a T -null-series. $\square$

Proposition 4.9. — There is an isomorphism of valued fields

$$\sigma: \mathbf{L}_p \xrightarrow{\cong} W(\overline{\mathbf{F}}_p)[p^{1/T}](t^{\mathbf{Q}})/\mathcal{N}_T.$$

Moreover, the following diagram commutes:

$$(b) \quad \begin{array}{ccc} W(\overline{\mathbf{F}}_p)(t^{\mathbf{Q}}) & \xhookrightarrow{\iota} & W(\overline{\mathbf{F}}_p)[p^{1/T}](t^{\mathbf{Q}}) \\ \downarrow \pi & & \downarrow \pi_T \\ \mathbf{L}_p & \xrightarrow[\sigma]{\cong} & W(\overline{\mathbf{F}}_p)[p^{1/T}](t^{\mathbf{Q}})/\mathcal{N}_T \end{array}.$$

Proof. — This is a direct consequence of the following standard fact in commutative algebra: given domains $R_1 \subset R_2$ and an ideal I of R_2 , if $R_1 + I = R_2$, then $R_1/(I \cap R_1) \cong R_2/I$, where the isomorphism is induced by the inclusion $R_1 \subset R_2$. $\square$

Remark 4.10. — For any element $f = \sum_{q \in \mathbf{Q}} [f(q)]p^q \in \mathbf{L}_p$, its lift $\hat{f} = \sum_{q \in \mathbf{Q}} [f(q)]t^q \in W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}}))$ embeds into $W(\overline{\mathbf{F}}_p)[p^{1/T}](t^{\mathbf{Q}})$ and then projects to an element of $W(\overline{\mathbf{F}}_p)[p^{1/T}](t^{\mathbf{Q}})/\mathcal{N}_T$, which can also be formally written as $\sum_{q \in \mathbf{Q}} [f(q)]p^q$. The commutative diagram in the proposition above shows that the isomorphism between $\mathbf{L}_p$ and $W(\overline{\mathbf{F}}_p)[p^{1/T}](t^{\mathbf{Q}})/\mathcal{N}_T$ is given by

$$\sum_{q \in \mathbf{Q}} [f(q)]p^q \in \mathbf{L}_p \mapsto \sum_{q \in \mathbf{Q}} [f(q)]p^q \in W(\overline{\mathbf{F}}_p)[p^{1/T}](t^{\mathbf{Q}})/\mathcal{N}_T.$$

5. Main theorem and its applications

Theorem 5.1. — Let $f = \sum_{s \in \mathbf{Q}} [f(s)]t^s \in \mathbf{L}_p^b$ be a p -adic Hahn series with $\text{Supp}(f) \subseteq \frac{1}{T_f} \mathbf{Z}[p^{-1}]$ for some integer $T_f \geq 1$. If there exists a sparse subset $S \subset \mathbb{P}$ such that $\|S\| := \{\|d\| \in \mathbf{Q} \mid d \in S\}$ is a set of representatives of $-T_f \cdot \text{Supp}(f)$ modulo $\mathbf{Z}$, then f is transcendental over $\mathbf{Q}_p$, and hence transcendental over $\mathbf{Q}_p$.

Proof. — We will use the notation of (b) freely throughout the proof without further explanation.

Suppose that f is a root of a polynomial $P(X) = a_n X^n + a_{n-1} X^{n-1} + \cdots + a_0 \in W(\overline{\mathbf{F}}_p)[X]$ with $a_n \neq 0$. By multiplying $P(X)$ by a suitable power of X , we may assume that S is (c, n) -sparse for some integer $c \geq 1$, where n equals the degree of $P(X)$. Thus there exist n elements $\underline{d}^{(1)}, \underline{d}^{(2)}, \dots, \underline{d}^{(n)} \in S$ satisfying condition (1b) of Definition 3.4 for n and c .

Since $\|S\|$ is a set of representatives of $-T_f \cdot \text{Supp}(f)$ modulo $\mathbf{Z}$, the set $-\frac{1}{T_f} \|S\|$ is a set of representatives of $\text{Supp}(f)$ modulo $\frac{1}{T_f} \mathbf{Z}$. For every $\underline{d} \in S$, we set

$$n_{\underline{d}} := -\frac{\|\underline{d}\|}{T} - \inf \left(\text{Supp}(f) \cap \left(-\frac{\|\underline{d}\|}{T} + \frac{1}{T} \mathbf{Z} \right) \right) \in \frac{1}{T} \mathbf{Z}.$$

This element is well defined because $\text{Supp}(f) \cap \left(-\frac{\|\underline{d}\|}{T} + \frac{1}{T} \mathbf{Z} \right)$ is non-empty by the assumption that $-\frac{1}{T_f} \|S\|$ is a set of representatives of $\text{Supp}(f)$ modulo $\frac{1}{T_f} \mathbf{Z}$, and is a subset of the well-ordered set $\text{Supp}(f)$, and hence has a minimum element. Then the set

$$\tilde{S} := \left\{ -\frac{\|\underline{d}\|}{T} - n_{\underline{d}} \mid \underline{d} \in S \right\} \subseteq \text{Supp}(f)$$

is a well-ordered set of representatives of $\text{Supp}(f)$ modulo $\frac{1}{T} \mathbf{Z}$, and is in bijection with S via the map

$$\mu: \underline{d} \mapsto -\frac{\|\underline{d}\|}{T} - n_{\underline{d}} = \inf \left(\text{Supp}(f) \cap \left(-\frac{\|\underline{d}\|}{T} + \frac{1}{T} \mathbf{Z} \right) \right).$$

Suppose that $f = \sum_{q \in \mathbf{Q}} [f(q)]p^q \in \mathbf{L}_p$. Then $\sigma(f) = \sum_{q \in \mathbf{Q}} [f(q)]p^q \in W(\overline{\mathbf{F}}_p)[p^{1/T}](t^{\mathbf{Q}})/\mathcal{N}_T$, and clearly $\text{Supp}(\sigma(f)) = \text{Supp}(f)$. For any $s \in \tilde{S}$, set $C_s := \sum_{w \in \mathbf{Z}} [f(s + \frac{w}{T})]p^{\frac{w}{T}} \in \check{\mathbf{Q}}_p(p^{1/T})$. By the construction of $\tilde{S}$, we have $C_s \neq 0$ and $f(s + \frac{w}{T}) = 0$ for every $s \in \tilde{S}$ and every integer $w < 0$. Thus $C_s = \sum_{w=0}^{\infty} [f(s + \frac{w}{T})]p^{\frac{w}{T}} \in W(\overline{\mathbf{F}}_p)[p^{1/T}]$, so that the element

$$\hat{f} := \sum_{s \in \tilde{S}} C_s \cdot t^s \in W(\overline{\mathbf{F}}_p)[p^{1/T}](t^{\mathbf{Q}})$$

is a lift of $\sigma(f)$ in $W(\overline{\mathbf{F}}_p)[p^{1/T}](t^{\mathbf{Q}})$.

By multinomial expansion, for any integer $i \geq 0$, one has

$$\widehat{f}^i = \sum_{q \in \mathbf{Q}} \left(\sum_{\substack{s_1, \dots, s_i \in \widetilde{S} \\ s_1 + \dots + s_i = q}} \prod_{k=1}^i C_{s_k} \right) t^q = \sum_{q \in \mathbf{Q}} \left(\sum_{\substack{\widetilde{\phi}: \widetilde{S} \rightarrow \mathbf{N} \\ \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) = i \\ \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \cdot s = q}} \frac{i!}{\prod_{s \in \widetilde{S}} \widetilde{\phi}(s)!} \prod_{s \in \widetilde{S}} C_s^{\widetilde{\phi}(s)} \right) t^q.$$

Since

$$\sigma \left(\sum_{i=0}^n a_i f^i \right) = \sum_{i=0}^n a_i \sigma(f)^i = 0$$

the element

$$\begin{aligned} \sum_{i=0}^n a_i \widehat{f}^i &= \sum_{q \in \mathbf{Q}} \left(\sum_{i=0}^n a_i \sum_{\substack{\widetilde{\phi}: \widetilde{S} \rightarrow \mathbf{N} \\ \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) = i \\ \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \cdot s = q}} \frac{i!}{\prod_{s \in \widetilde{S}} \widetilde{\phi}(s)!} \prod_{s \in \widetilde{S}} C_s^{\widetilde{\phi}(s)} \right) t^q \\ &= \sum_{q \in \mathbf{Q}} \left(\sum_{\substack{\widetilde{\phi}: \widetilde{S} \rightarrow \mathbf{N} \\ \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \leq n \\ \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \cdot s = q}} a_{\sum_{s \in \widetilde{S}} \widetilde{\phi}(s)} \frac{\left(\sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \right)!}{\prod_{s \in \widetilde{S}} \widetilde{\phi}(s)!} \prod_{s \in \widetilde{S}} C_s^{\widetilde{\phi}(s)} \right) t^q \end{aligned}$$

is a T -null-series in $W(\overline{\mathbf{F}}_p)[p^{1/T}][t^{\mathbf{Q}}]$, i.e.

$$\sum_{w \in \mathbf{Z}} p^w \sum_{\substack{\widetilde{\phi}: \widetilde{S} \rightarrow \mathbf{N} \\ \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \leq n \\ \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \cdot s = q + \frac{w}{T}}} a_{\sum_{s \in \widetilde{S}} \widetilde{\phi}(s)} \frac{\left(\sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \right)!}{\prod_{s \in \widetilde{S}} \widetilde{\phi}(s)!} \prod_{s \in \widetilde{S}} C_s^{\widetilde{\phi}(s)} = 0$$

for every $q \in \mathbf{Q}$. We specialize this at $q = -\frac{r_0}{T}$, where

$$r_0 := \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}),$$

to get

$$(c) \quad \sum_{w \in \mathbf{Z}} p^w \sum_{\substack{\widetilde{\phi}: \widetilde{S} \rightarrow \mathbf{N} \\ \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \leq n \\ \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \cdot s = -\frac{r_0}{T} + \frac{w}{T}}} a_{\sum_{s \in \widetilde{S}} \widetilde{\phi}(s)} \frac{\left(\sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \right)!}{\prod_{s \in \widetilde{S}} \widetilde{\phi}(s)!} \prod_{s \in \widetilde{S}} C_s^{\widetilde{\phi}(s)} = 0.$$

For every $\widetilde{\phi}: \widetilde{S} \rightarrow \mathbf{N}$ such that $\sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \leq n$ and $\sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \cdot s \equiv -\frac{r_0}{T} \pmod{\frac{1}{T}\mathbf{Z}}$, we set $\phi := \widetilde{\phi} \circ \mu$. Then

$$\sum_{\underline{d} \in S} \phi(\underline{d}) = \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \leq n$$

and

$$\begin{aligned} \sum_{\underline{d} \in S} \phi(\underline{d}) \cdot \|\underline{d}\| &= -T \sum_{s \in \tilde{S}} \tilde{\phi}(s) \cdot s - \sum_{\underline{d} \in S} \tilde{\phi}\left(-\frac{\|\underline{d}\|}{T} - n_{\underline{d}}\right) (T \cdot n_{\underline{d}}) \\ &\equiv -T \sum_{s \in \tilde{S}} \tilde{\phi}(s) \cdot s \equiv r_0 \pmod{\mathbf{Z}}. \end{aligned}$$

By the claim, one knows that $\phi = \phi_0$, i.e. $\tilde{\phi} = \phi_0 \circ \mu^{-1}$. As a result, one can rewrite (c) as

$$p^T \sum_{\underline{d} \in S} \phi_0(\underline{d}) \cdot \mu(\underline{d}) + r_0 a_n \frac{n!}{\prod_{\underline{d} \in S} \phi_0(\underline{d})!} \prod_{\underline{d} \in S} C_{\mu(\underline{d})}^{\phi_0(\underline{d})} = 0,$$

implying that $a_n = 0$ or $C_{\mu(\underline{d})} = 0$ for some $\underline{d} \in S$. Both cases lead to contradictions. The result follows. $\square$

Remark 5.2. — In the above proof, the blue sums are in fact finite by [Poo93, Lemma 1] (Poonen1993.pdf), while the red sums and products are well-defined because the function inside is finitely supported by the definition of $\tilde{\phi}$.

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